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import card_sets
import random
import os
import sys

# Display all values inside a tuple nested in a dictionary, allowing
the user to go through the flashcards.
def display_cards(input_dict):
    for key, value in input_dict.items():
        print(f"{key}. {value[0]}")
        input()
        print(f"{value[1]}\n\n\n")

    while True:
        play_again_input = input("Finished the set! Play it again?
[Y/n]")
        if play_again_input == 'n':
            return False
        elif play_again_input in ('Y', 'y', ''):
            return True

# Shuffles the terms in a card
# 1. Assigns the values of a dictionary to a temporary variable
# 2. Shuffles the list using random.shuffle
# 3. Uses the zip function to combine the shuffled values with the
original keys, allowing for the questions and answers to stay paired
but for the order to be shuffled.
def shuffle_terms(card_set):
    original_values = list(card_set.values())
    random.shuffle(original_values)
    return(dict(zip(card_set, original_values)))

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# Shuffler logic to allow for the program to run multiple times.
# Works by prompting the user, then returning the current_flashcards
that are either shuffled or unshuffled, depending on the user's
input.
def shuffler(input_set):
    while True:
        shuffle_determine = input("Would you like to shuffle the
cards? [Y/n]")
        if shuffle_determine == 'n':
            print("\nCards won't be shuffled.")
            current_flashcard = input_set
            break
        elif shuffle_determine in ('Y', 'y', ''):
            print("\nCards will be shuffled.\n")
            buffer = shuffle_terms(input_set)
            current_flashcard = buffer
            break
    return current_flashcard

# Main program code
def main():
    print("\nWelcome to CSP Cards, an AP CSP MCQ Exam Practice tool
written in Python!\nTo begin, select a flashcard set number:\n")
    # Lists out all flashcards in the 'db' variable.
    for index, value in enumerate(card_sets.db):
        print(f"{index + 1}: {value[0]}")
    print("\n0: Exit")

    while True:
        # Input validation for every possible case.
        try:
            user_selection = int(input())
            # Assigns the flashcard selected to a new variable for
ease-of-accesss.

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        current_flashcard = card_sets.db[user_selection - 1][1]
        if user_selection == 0:
            # Allows user to exit the program without using a
            keyboard interrupt (Ctrl+C) or an EOF (Ctrl+D).
            sys.exit()

        break
    except ValueError:
        print("Please enter a single number!\n")
    except IndexError:
        print("Please select a flashcard number in the
database!\n")
    print(f'You have selected "{card_sets.db[user_selection -
1][0]}"')

    # Shuffles flashcards using a buffer.
    # This method allows the user to be asked every single time.
    buffer = shuffler(current_flashcard)
    current_flashcard = buffer
    # Dummy code to test:
    # print(type(current_flashcard))
    print("Let's play!\n")

    # Prints out flashcards using display_cards().
    # display_cards() returns the user's play again input, so this
    loop determines whether the flashcard set is repeated or if the
    program starts again.
    # If the user selects to play again, the shuffler runs again to
    determine if the user still wants to shuffle.
    # The newly shuffled or unshuffled cards are reassigned to
    current_flashcard, and the display_cards() function runs again.
    # If the user decides not to play again, the entire main()
    function runs again, and the screen is cleared.
    while True:
        if display_cards(current_flashcard):
            print()

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        buffer = shuffler(card_sets.db[user_selection - 1][1])
        current_flashcard = buffer
    else:
        print()
        return 0

if __name__ == '__main__':
    # Using this convention prevents the main() function from running
    # if this file is imported.
    # This allows for scalability such as implementing a GUI mode.
    while True:
        # Clear screen, command obtained from StackOverflow:
        #
https://stackoverflow.com/questions/2084508/clear-the-terminal-in-python
        os.system('cls' if os.name == 'nt' else 'clear')
        # Allows main code to be looped through the use of return
        # statements.
        main()

# All flashcard information was generated by Gemini 3 Pro.
# No code was generated by Generative AI.

flashcard_set_1 = {
    1: ("A code segment assigns  $a \leftarrow 10$  and  $b \leftarrow 20$ . It then executes  $a \leftarrow b$  followed by  $b \leftarrow a$ . What are the final values of  $a$  and  $b$ ?", "a = 20, b = 20. (The original value of  $a$  was overwritten in the first step)."),
    2: ("Which of the following best describes an undecidable problem?", "A problem for which no algorithm can be constructed that is always capable of providing a correct yes or no answer."),
    3: ("A list numbers contains [10, 20, 30, 40]. What is displayed by the code: DISPLAY(numbers[2] + numbers[4])?", "60 (AP CSP lists are 1-indexed, so index 2 is 20 and index 4 is 40)."),

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4: ("A binary search is performed on a sorted list of 1,000
elements. What is the approximate maximum number of comparisons
needed?", "10 or 11 (Since  $2^{10} = 1024$ , it takes about 10 steps to
halve the list down to one item)."),

5: ("What is the primary difference between a simulation and a
real-world experiment?", "Simulations allow testing of hypotheses
that might be too dangerous, expensive, or time-consuming to perform
in the real world."),

6: ("Which of the following is true about heuristics in
programming?", "They provide a solution that is good enough
(approximate) when an exact solution is unreasonable to compute."),

7: ("If a robot uses the logic REPEAT UNTIL (GoalReached) to
navigate a maze, what type of iteration is this?", "Indefinite
Iteration (The number of steps is not known when the loop starts)."),

8: ("Which feature of a procedure (function) helps most with code
reuse and reducing complexity?", "Parameters (They allow the
procedure to generalize its behavior for different input values)."),

9: ("A program checks IF (grade > 90). If this is false, it checks
ELSE IF (grade > 80). If a student has an 85, which block runs?",
"Only the ELSE IF (grade > 80) block runs."),

10: ("Two boolean variables, isValid and isReady, are both FALSE.
What does (NOT isValid) AND (NOT isReady) evaluate to?", "TRUE. (NOT
False = True. So it becomes True AND True)."),
}

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flashcard_set_2 = {

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1: ("How does the Internet handle a file sent from one computer to
another?", "The file is broken into packets, which may take different
paths and arrive out of order, and are reassembled by the receiving
device."),

2: ("What feature allows the Internet to continue operating
properly even if a specific connection or router fails?", "Fault
Tolerance (achieved through Redundancy)."),

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3: ("Which of the following best describes the Digital Divide?",
"Differences in access to computing devices and the Internet based on
socioeconomic, geographic, or demographic characteristics."),
4: ("What is the main function of the Transmission Control
Protocol (TCP)?", "It manages the reliable transmission of data,
including error-checking and requesting the re-transmission of
missing packets."),
5: ("Why does the Internet use open, non-proprietary standards
(like HTTP and IP)?", "To ensure interoperability, allowing devices
from different manufacturers to communicate without restriction."),
6: ("In public key encryption, if User A wants to send a secure
message to User B, which key does User A use to encrypt the
message?", "User B's Public Key. (Only User B's Private Key can then
decrypt it)."),
7: ("A system adds more servers to handle increased traffic
without noticing a drop in performance. What property is this?",
"Scalability."),
8: ("What is a Distributed Denial of Service (DDoS) attack?", "An
attempt to overwhelm a server with traffic from many different
sources to make it unavailable to legitimate users."),
9: ("What is the relationship between bandwidth and latency?",
"Bandwidth is the amount of data that can be sent in a fixed time;
Latency is the time it takes for a single piece of data to travel."),
10: ("What is the purpose of a Digital Certificate?", "It verifies
that a public key belongs to a specific entity, preventing impostors
from intercepting secure data."),
}

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flashcard_set_3 = {
1: ("A social media site uses an algorithm to populate user feeds.
Why might this algorithm inadvertently create echo chambers?", "The
algorithm is optimized to maximize engagement, so it filters out
opposing viewpoints to show users content that aligns with their
beliefs."),

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2: ("Which Creative Commons license allows others to use a work,
but only if they give credit and do not make money from it?", "CC
BY-NC (Attribution-NonCommercial)."),
3: ("What is Crowdsourcing?", "Obtaining information, work, or
funding from a large group of people via the Internet."),
4: ("A photo sharing app stores metadata with every picture. What
is an example of a privacy risk associated with metadata?", "The
metadata might contain the GPS coordinates of where the photo was
taken, revealing the user's home location."),
5: ("What is the most likely source of bias in a Machine Learning
algorithm?", "The data used to train the algorithm was not
representative of the entire population."),
6: ("What is Phishing?", "A social engineering attack where a bad
actor sends a deceptive email to trick a user into revealing a
password."),
7: ("What is the Citizen Science model?", "Scientific research
where data collection or analysis is performed by members of the
general public."),
8: ("What is the difference between Open Source and Freeware?",
"Open Source makes the actual source code available for inspection;
Freeware is free to use but the code is hidden."),
9: ("A user logs into a site using a password and a code sent to
their phone. What is this called?", "Multifactor Authentication
(MFA)."),
10: ("Why is it difficult to legally protect software algorithms
under copyright law compared to books or music?", "Copyright protects
the expression of an idea (the specific code), but not the functional
logic or math behind the algorithm itself."),
}

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db = [
    ("Algorithms & Programming", flashcard_set_1),
    ("Computing Systems & Networks", flashcard_set_2),
    ("Impact of Computing", flashcard_set_3),

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